This year the goal of the OR project is to build a well-performing financial portfolio.

To get the ball running, your goal for the first weeks is to build such a portfolio out the BEL20 companies using the Markowitz model of risk. A short report of max 5 pages describing your methodology and findings is due on the 8th of October 2025, to be submitted on Moodle.

The goal on this first assignment is to :

- Use the data provided in the BEL20.zip file on Moodle to compute the average returns and the variance-covariance matrix of the returns of the 21 assets.
- Use this as the input data to set up an optimization model to compute the optimal portfolio under the Markowitz model of risk.
- Use that optimization model to compute:
  1) the efficient frontier assuming no short selling, and
  2) the efficient/tangent portfolio when short selling is allowed.
- As a bonus, try to construct a good portfolio that contains only 5 assets (no short selling).